INTEGRATING ERC-20 ALLOWANCE EDGES INTO PAGERANK FOR ENHANCED ON-CHAIN WALLET REPUTATION SCORING

CHAPTER 1: Introduction and Research Problem

by

TAEHONG KWON

Student number: 28576810

submitted in accordance with the requirements for the degree of

DOCTOR OF PHILOSOPHY

in the subject

COMPUTER SCIENCE

at the

UNIVERSITY OF SOUTH AFRICA

SUPERVISOR: Prof. Ernest Mnkandla

CO-SUPERVISOR: Prof. Donatien Koulla Moulla

CO-SUPERVISOR: Dr. David Sena Attipoe

October 2026

1 Introduction and Research Problem

Decentralized Finance (DeFi) has transformed how capital is allocated on public blockchains: lending, borrowing, and leveraged trading can now run without traditional intermediaries (Schär, 2021). Most credit-like transactions are still heavily collateralized, though, because protocols cannot draw on off-chain identity frameworks or existing credit data (Harvey et al., 2021). Under pseudonymity, the central issue is how to allocate capital efficiently among unknown parties.

This thesis asks what ERC-20 allowance edges add to PageRank as a measure of on-chain wallet reputation. The allowance edges come from the `Approval` events that `approve` and `permit` emit, and EndorseRank is PageRank computed on them with the edge weights of the transfer-based Adaptive Weighted PageRank (AWP) of T. Do et al. (2023). Both are run on the same Arbitrum One contract accounts, with a coupled walk over both edge types and the raw degrees that the walks smooth, and evaluated by the rank-correlation method used with AWP (T. Do et al., 2023), extended with paired contrasts, raw-degree baselines and temporal holdouts. In this thesis, an on-chain reputation metric is a value derived from on-chain activity and interactions without any off-chain identity information (H.-D. Do & Do, 2023).

1.1 Background: DeFi, collateralization, and pseudonymity

Public blockchains make programmable settlement possible without a central authority (Nakamoto, 2008). A smart contract can encode the terms of a deal (Szabo, 1997), and a DeFi platform that uses such contracts clears trades on the blockchain instead of routing them through a custodian (Buterin, 2014). Protocols then stack these contracts into decentralized markets for lending, automated market making and perpetual futures (Werner et al., 2022). On Arbitrum One and other L2 networks, fees are low enough that ordinary users transact often, and every transfer, approval and

protocol event can be inspected wallet by wallet.

Even with this visibility, DeFi credit markets remain inherently over-collateralized. Most protocols make a borrower or trader post more collateral than the position is worth, because the counterparty is a pseudonymous account whose owner has not been verified (Gudgeon, Perez, et al., 2020; Harvey et al., 2021). Pseudonymity is a basic property of public ledgers. Anyone can create addresses at almost no cost, and one entity can own more than one wallet (Douceur, 2002). Conventional credit ratings depend on personal histories that translate poorly to this setting (Packin & Lev-Aretz, 2024), so protocols control risk through collateral reserves, liquidation mechanisms, and protocol rules instead of personalized reputation.

Collateralization protects protocols, but it uses capital inefficiently. Traders and borrowers have to tie up funds they could otherwise invest, and the platforms offer none of the under-collateralized products that traditional finance extends to people with a credit record. Researchers have therefore turned to on-chain signals as a substitute for off-chain credit histories: transfer-graph centrality (H.-D. Do & Do, 2023), network risk propagation (D. Lin et al., 2025), prospect theory and blockchain-based credit scoring (Hassija et al., 2020), and machine learning pipelines over lending protocols (Kotb, 2023). Each of these lines of work assumes that public ledger data contains information usable for risk assessment without deanonymizing users.

Graph ranking algorithms appear well suited to this problem. PageRank and its variants rank nodes by recursive endorsement, so that a node's score depends on the endorsements it receives from other nodes (Page et al., 1999). Earlier web trust algorithms such as EigenTrust (Kamvar et al., 2003) and TrustRank (Gyöngyi et al., 2004) apply similar ideas to peer-to-peer trust and to web spam. H.-D. Do and Do (2023) use both PageRank and HodgeRank on a graph of transfers to judge how trustworthy Ethereum users are, and call the result a form of social credit. AWP (T. Do et al., 2023) is another transfer-graph ranking algorithm. It weights each transfer by a bounded function of its value and a logistic function of its age, and it restarts the walk at addresses that sent recently; its authors validate it by showing that it correlates with inbound transfer count and value, taken as proxies for economic importance. These studies establish transfer-graph reputation as a credible

baseline, but they leave out explicit spending authorization, a separate on-chain action recorded as ERC-20 allowances (Lundfall, 2020; Vogelsteller & Buterin, 2015).

1.2 Motivation for on-chain reputation

Four main factors drive the need for on-chain reputation research in leveraged DeFi.

- Capital efficiency: protocols are seeking ordinal ranking signals that could support tiered services (more frequent limit increases, lower fees, or higher priority in monitoring, for example), and perhaps even differentiated collateral requirements in future controlled experiments, while settlement stays on chain, where the contract enforces collateralization ratios and liquidation conditions (Schär, 2021).
- Auditability: off-chain credit scores are opaque, but graph-based scores computed from public event data can be recomputed and audited, which would help to ease some of the concerns about black-box decentralized credit scoring (Packin & Lev-Aretz, 2024).
- Operational cost: a score that is refreshed often, before each leverage adjustment for instance, should be cheap to recompute.
- Standardization: other markets use screening ratios to tell counterparties apart, but pseudonymous DeFi has no standard ordinal language for doing so at the wallet level. Chapter 5 discusses where the scores studied here could serve that role and where they cannot.

Transfer-based AWP is public and auditable, but its graph grows with every payment (T. Do et al., 2023). Allowance graphs are a structurally different alternative. Whenever a user approves an ERC-20 token, the contract records how much of that token a spender may transfer (Lundfall, 2020; Vogelsteller & Buterin, 2015), and each new approval supersedes the older ones, so the latest approval of each owner–spender pair and token stands for the history of that pair. Approvals are also rarer than transfers. On the same contracts the endorsement graph G_{approve} therefore has fewer than half as many edges as the transfer graph G_{transfer}, and the PageRank solve, whose cost grows linearly with $|E|$, takes correspondingly less time (Table 4.1). The same solver runs in both cases,

so the speedup comes from the sparser graph and is not an algorithmic improvement; Claim C1 treats it as an operational property of the representation.

Efficiency aside, allowances also mean something different. A transfer might reflect routing, market making, or a passive receipt, whereas an allowance is an active grant of permission to spend funds. Treating allowances as endorsement edges carries PageRank's citation reading (Page et al., 1999) over to a trust relationship that protocols already depend on. Whether this produces different rankings, and whether those rankings agree differently with authorization and transfer checks, is examined on a matched cohort of $n = 15{,}107$ contract accounts on Arbitrum One.[1]

1.3 Research problem

AWP infers trust from transfer history through weighted sums and logistic time-decay (T. Do et al., 2023). It (i) must aggregate the whole transfer history, (ii) models a high-value wallet as one that receives many inbound transfers, and (iii) assigns no role to spending authorization. EndorseRank, the score studied here, keeps AWP's edge weights and solver but builds the directed graph from the latest allowance of each owner–spender pair and token. The resulting graph is sparser, encodes a different signal and ranks a different side of each relationship. An allowance points from the owner to the spender, so EndorseRank scores the spenders that receive approvals, such as routers and vaults, and gives every owner that receives none the same value.

The main research question is: what do ERC-20 allowance edges add to PageRank as a reputation score for contract accounts on Arbitrum One, compared with the transfer walk and with the raw degrees that both walks smooth? Research questions RQ1–RQ5 (Section 1.6) break it down. Because both edge types live on the same addresses, a single walk can also mix them node by node with a fixed weight λ, which defines a Coupled PageRank (C-PR); at $\lambda = 1$ and $\lambda = 0$ it walks one layer alone. The mixture is worth testing because the two edge types record different acts on the same

[1] Matched means that every method scores the same contracts: those approved by at least three distinct owners on WETH, USDC, USDT0, ARB or WBTC between 1 October 2023 and 30 June 2026, a list drawn from the approval logs alone (Section 3.2).

contracts: an operator that already runs a transfer walk could add allowance edges to it instead of running a second score.

Agreement with statistics of a score's own edges, or prediction of more of the same edges, does not yet show that a score measures reputation in a substantive sense. A comparison registered with the second holdout window, which adds no research question, therefore also uses an outcome built from neither edge type and observed after the freeze: the later liquidation avoidance of the GMX V2 accounts that are contracts, in small samples (Section 3.7.3); it is applied to the revised scores as a re-analysis.

The scores appear in Table 1.1. Table 1.2 lists hypotheses (H1–H3), research questions (RQ1–RQ5), and empirical claims (C1–C6) from Chapter 4.

Table 1.1: The scores evaluated in this thesis (matched contract cohort, $n = 15{,}107$).

Score	Input graph	Role
EndorseRank	Latest ERC-20 allowances of five tokens in US dollars (G_{approve}), weighted as in AWP	Allowance delegation; the object of study in this research
AWP	Transfers of five tokens in US dollars (G_{transfer})	Transfer flow; the published transfer-graph method of T. Do et al. (2023) (Section 3.6.1)
C-PR	Both layers, US-dollar amounts with each allowance capped, mixed per node with weight λ	Coupled operator used to test whether the two edge types combine (RQ5). Its ablation is S-PR, a transfer walk restarted from the allowance walk
Raw degrees	G_{approve} and G_{transfer}, counted per node	In-approve degree and transfer in-degree, the local counts that the PageRanks smooth: proxies in the same window and baselines out of window

The main terms are defined as follows:

- *DeFi (Decentralized Finance):* Financial services such as lending and borrowing, implemented as smart contracts on public blockchains and generally run without a central operator in charge of custody or settlement (Schär, 2021).
- *Smart contract:* Self-executing program code that runs on a blockchain and carries out an agreement without a trusted intermediary (Szabo, 1997).

- *Wallet / Address:* A blockchain account that holds funds and interacts with smart contracts, identified by its address (Buterin, 2014). It is either an externally owned account (EOA), a user's wallet controlled by a private key, or a contract account governed by its code (Antonopoulos & Wood, 2018). The wallets ranked here are contract accounts; EOAs enter only as owners, senders and recipients. No off-chain identity is assumed or recovered.
- *On-chain reputation score:* A number computed from on-chain data that expresses a wallet's relative position for financial activity under pseudonymity (H.-D. Do & Do, 2023). Here the scored addresses are contracts, and the only off-chain inputs are the five-token list and daily US-dollar prices. Scores are ordinal and meant for ranking; their absolute scale has no meaning for credit limits.
- *PageRank:* A graph ranking algorithm that spreads influence through a directed graph with a damping factor (Page et al., 1999). Adaptive Weighted PageRank (AWP) is PageRank on the transfer graph, with edges weighted by the value and age of each transfer and restarts at recently active senders (T. Do et al., 2023); here values are in US dollars, and the value weight is one half at the median amount. EndorseRank applies AWP's edge weights to the latest-allowance graph and restarts uniformly (Section 3.6).
- *Coupled PageRank (C-PR):* One random walk over both the allowance and the transfer layer, which at each wallet mixes the two rows with weights λ and $1 - \lambda$ over the layers that wallet occupies (Section 3.6.3). Endorsement-seeded PageRank (S-PR) keeps the transfer walk but restarts it from the allowance walk, as TrustRank (Gyöngyi et al., 2004) restarts from a seed set.
- *ERC-20 allowance / Approve / Permit:* Contract methods that record how many of the owner's tokens a spender may spend on the owner's behalf (Lundfall, 2020; Vogelsteller & Buterin, 2015). `Approve` is an ordinary on-chain call; EIP-2612 `permit` makes the same state change through a signature. For each owner–spender pair and token, EndorseRank uses the latest non-zero allowance.

- *Endorsement graph:* A directed graph whose edges denote an explicit delegation of spending authority, as opposed to an actual transfer of assets (Vogelsteller & Buterin, 2015). An edge $u \to v$ means that the owner u, an EOA or a contract, has approved v to spend tokens belonging to u.
- *Domain alignment:* The degree to which a reputation ranking agrees with an observable proxy ranking, measured with Spearman's ρ (Spearman, 1904) and Kendall's τ (Kendall, 1938) and read as evidence on construct validity (Cronbach & Meehl, 1955); with proxies built from a score's own edges it is only an intended-construct check.

1.4 Temporal holdout

The contemporaneous allowance Kendall τ (Table 4.3) is an intended-construct check: PageRank on G_{approve} should agree with spender-side approval degree, because it smooths that degree globally. To test whether the score conveys information beyond the same-window graph, the scores are frozen at t_1 (31 March 2026, 23:59:59 UTC) and compared with two labels that only become available in April–June 2026: new owner–spender approval pairs (the endorsement construct) and new transfer senders (the flow construct). In this first window, W0, the holdout cohort ($n = 17{,}050$) consists of every contract spender that holds at least one positive latest allowance of one of the five tokens at t_1 in the extracted approval data. Each PageRank is also compared with the raw degree it smooths at t_1.

A second window, W1, was registered at 16:49 UTC on 9 October 2026, after the W0 results were known and before the study extracted any log dated after 30 June 2026. Its scores are frozen on 30 June 2026 and its labels counted from July to September 2026. It repeats the two spender labels on the $n = 18{,}152$ contract spenders of that freeze and adds one for the GMX V2 traders that are contracts ($n = 70$): the share of their closes that did not end in liquidation (Section 3.7.2).

The analysis plan of 9 October 2026 and the registration of W1 define the registered analysis, with amounts in raw base units of every token. After its results were known, the amounts were revised

on 10 October 2026 to US dollars on five widely traded tokens (Section 3.5.1). The revised analysis is the main one; its verdicts under the rules of the registered analysis are a re-analysis, not registered tests, and the registered analysis reached the same verdicts (Appendix 8.6).

1.5 Research objectives

The research has five objectives, one for each of RQ1–RQ5. Each is stated with the measurement used to operationalize it and with the criterion that Chapter 5 applies when judging whether the objective was met.

O1. Establish whether EndorseRank and AWP order the contracts of the same cohort differently (RQ1). Measurement: Kendall τ between the two score vectors ($n = 15{,}107$) with a 95% bootstrap interval. Criterion: the interval contains neither 0 (unrelated orderings) nor 1 (identical orderings).

O2. Assess how closely each score agrees with each construct in the same window (RQ2). Measurement: family-level Kendall τ with 95% intervals for every score and construct. Criterion: a score agrees with a construct when its interval on that family excludes 0; agreement with the construct of the score's own edges is read as an intended-construct check, because those proxies come from the same events as the graph.

O3. Measure what each score costs to compute on the same contracts, and how runtime responds to the number of sampled contracts (RQ3). Measurement: solve time (mean and standard deviation of five timed runs after one warm-up run), memory, iterations, $|V|$ and $|E|$, all under one protocol. Criterion: the time ratio is reported only together with the ratio of $|E|$, so that any difference can be traced to the size of the graph.

O4. Test whether scores frozen at t_1 predict new approvals and new transfer senders at t_2 among the cohort's spenders, and whether any PageRank does better than the raw degree it smooths (RQ4). Measurement: Kendall τ with 95% intervals between every score computed at the freeze, the two raw degrees included, and both labels, in W0 and W1, with paired contrasts

Table 1.2: Summary of hypotheses, research questions, and empirical claims.

Item	Addresses	Reported in
H1 / RQ1	Do EndorseRank and AWP produce different orderings of the same contracts?	§4.4; Table 4.5; Claim C3
H2 / RQ2	How closely does each score agree with the allowance checks and with the transfer checks?	Tables 4.3 and 4.5; Claim C2
H3 / RQ3	What does each score cost to compute at equal n?	Tables 4.1 and 4.2; Claim C1
RQ4	Do t_1 scores predict t_2 new approvals and new senders, and does any PageRank do better than the raw degree it smooths?	§4.6; Tables 4.10, 4.11, 4.12, 4.13, 4.14 and 4.16; Claim C4
RQ5	Does the coupled operator C-PR improve on a walk over the transfers alone, and does it rank contracts better than a walk over either layer alone?	§4.3.1 and §4.6; Tables 4.4, 4.11, 4.14 and 4.15; Claim C5
C1	Runtime and graph size of each score: speed follows graph size, not the algorithm; C-PR and S-PR are slower than AWP	Table 4.1
C2	Same-window agreement of each score with the checks of its own edge type, partly built in	Tables 4.3 and 4.5
C3	EndorseRank and AWP order the cohort differently	Table 4.5; Figure 4.3
C4	Out of window, AWP ranks later new senders above the transfer in-degree in both windows, while EndorseRank ranks later new approval pairs below the in-approve degree in both (contrasts computed after the labels were known)	Tables 4.10, 4.11, 4.12 and 4.16
C5	C-PR does not beat its two layers walked alone: rule A fails and rule B is not met (gain on new approval pairs; non-inferiority fails on new senders and narrowly on liquidation-free closes); against AWP it gains on new approval pairs through the spenders AWP scores zero and loses on new senders; allowance edges help mainly when they carry the walk	Tables 4.3, 4.4, 4.11, 4.15 and 4.20 (post hoc); §4.6.2
C6	On later liquidation avoidance of GMX V2 contract traders, no count-level lead of allowance over transfer in the registered window (positive in W0); PageRank versions not resolved; approval count lower on the profitable and non-loss shares in W1; small samples	Tables 4.18 and 4.19

of each PageRank against its own degree. Criterion: a score predicts a label when its interval excludes 0, and a PageRank improves on its degree only when the interval of the paired difference lies above 0. Neither the degree contrasts nor this criterion was fixed before the W0 labels were known.

O5. Test whether one walk across both edge types improves on a walk over the transfers alone, and whether it beats a walk over either layer alone (RQ5). Measurement: C-PR at $\lambda = 0.5$ (with 0.25 and 0.75 as sensitivity values) and S-PR, compared through paired intervals on $\Delta\tau$ with the two layers walked alone and with EndorseRank and AWP, on the holdout and same-window labels. Criteria: (a) rule A, fixed in the analysis plan of 9 October 2026 before any log of the study was extracted, under which C-PR does no worse than the allowance walk on new approvals and better than the transfer walk on new senders in W0, and in the same window lies within the interval of the leading layer on the transfer and allowance families and above both layers on Sybil stability; (b) rule B, registered on 9 October 2026 after the W0 results were known and before the study extracted the W1 logs, under which C-PR beats the transfer walk on new approval pairs in W1 and is not worse by more than 0.01 on new senders or on the traders' liquidation-free close rate. Both rules are applied to the revised scores as a re-analysis. A registered sensitivity analysis repeats (b) with AWP in place of the transfer walk; it does not change the rule.

1.6 Research questions

Three hypotheses go with the first three research questions. H1: on the same set of contracts, EndorseRank and AWP produce different orderings. H2: each score agrees with the checks built from the edges it walks on. H3: at the same value of n, a score's computation time grows with the size of its graph, so EndorseRank, on the sparsest graph, is the fastest to compute. H1–H3 hold largely by construction and act as checks of the instruments. The falsifiable tests are RQ4 and RQ5, the latter answered by rules fixed before the study extracted the relevant data and re-applied to the

revised scores (Section 3.7).

RQ1. On the same set of Arbitrum contracts, do EndorseRank and AWP produce different orderings?

RQ2. How closely does each score agree with current allowance checks and with current transfer checks (Spearman's ρ and Kendall's τ)?

RQ3. What does each score cost to compute on the same set of contracts, as measured by time, memory, and iterations?

RQ4. Given scores computed at time t_1, do the scores predict t_2 new owner–spender approvals to inbound spenders and new transfer senders, and does any PageRank predict them better than the raw degree it smooths at t_1?

RQ5. Does a coupled PageRank running on both the allowance and the transfer layer improve on a walk over the transfers alone, such as AWP, and does it rank contracts better than a walk over either layer alone, out of sample as well as in sample?

RQ4 uses later labels of the same two constructs because no default or repayment label exists for the contracts ranked here (Section 2.8). RQ5 moves from comparing the two edge types to combining them; its rule for improving on the transfer walk, rule B, came after the W0 labels had been seen, so it is judged only on W1 (Sections 3.7 and 3.7.2).

1.7 Summary of the literature review

Chapter 2 reviews four areas. In graph ranking, PageRank (Page et al., 1999) ranks nodes by recursively propagated endorsement at a per-iteration cost linear in the number of edges (Langville & Meyer, 2006), and EigenTrust (Kamvar et al., 2003) and TrustRank (Gyöngyi et al., 2004) adapt it to peer-to-peer trust and web spam with normalization and seed sets. In on-chain reputation, H.-D. Do and Do (2023) apply PageRank and HodgeRank to Ethereum transactions, and AWP (T. Do et al., 2023) weights transaction edges by value and age and checks its ranking against inbound transfer count and value, statistics drawn from the same data as its graph. D. Lin et al. (2025) prop-

agate account risk through the transaction network, and surveys of blockchain trust and reputation systems (Bellini et al., 2020) and of reputation systems built on smart contracts (Almasoud et al., 2020) describe designs that rest on transfers or explicit ratings. Deployed DeFi tooling such as Human Passport instead aggregates on-chain and off-chain credentials to decide whether an address belongs to a distinct person (Human Passport, n.d.). None of this work takes the ERC-20 allowance (Vogelsteller & Buterin, 2015) as a ranking edge; address clustering uses approvals only to link addresses of one entity (Victor, 2020). In DeFi risk and measurement, systematizations classify risk by protocol layer (Werner et al., 2022) and by the attacks seen in practice (Zhou et al., 2023), and wash trading (Cong et al., 2023; Victor & Weintraud, 2021), extractable value (Qin et al., 2022), scam tokens (Xia et al., 2021) and airdrop farming (Luo et al., 2025) show that much transfer activity is artificial. In measurement theory, construct validity (Cronbach & Meehl, 1955; Messick, 1995) separates what a score measures from external criteria, and the bootstrap (DiCiccio & Efron, 1996; Efron, 1979) gives distribution-free intervals for rank statistics. The gap (Section 2.21) is an allowance-edge ranking evaluated on one cohort beside the transfer method and the raw degrees, with confidence intervals, paired differences, a fixed timing protocol and temporal holdouts.

1.8 Summary of the research methodology

Chapter 3 applies a quantitative, comparative design to public ledger data. ERC-20 `Approval` and `Transfer` logs on Arbitrum One from 1 October 2023 to 30 September 2026 are extracted from Google BigQuery in two scans, the second after W1 was registered. The analysis keeps the events of five tokens around the matched cohort ($n = 15{,}107$), valued in US dollars (Section 3.5.1). One power-iteration PageRank implementation scores the latest-allowance graph and the transfer graph, and the coupled operator runs on the same solver. Each score is compared with Spearman's ρ and Kendall's τ against same-window transfer, allowance and Sybil-stability proxies and, out of window, against new approval pairs and new transfer senders on the spender contracts of W0 and W1, with the raw degrees at each freeze as baselines. Except in the value-slope and sample-

definition checks, every coefficient has a 95% interval from 400 paired contract resamples (seed 42), and method comparisons are paired. A comparison registered with W1 sets allowance-side against transfer-side scores on the later liquidation avoidance of GMX V2 contract traders in both windows (2,000 paired resamples; Bonferroni 97.5% intervals for its two decision contrasts). The code, configuration and machine-readable summaries behind every table are in a public repository (Section 3.13), and the study uses only public data (Section 3.16).

1.9 Contributions

The thesis makes five contributions to the study of on-chain reputation and DeFi risk. EndorseRank and AWP both apply weighted PageRank (Page et al., 1999) to a single edge type, each with its own restart distribution; only the coupled operator changes the transition rule. What is new is therefore an edge type, an evaluation design, an analytic account of what the allowance walk computes on this sample, and a test of whether the two edge types combine.

- Latest-allowance edge construction (EndorseRank): an endorsement graph in which an edge $u \to v$ stands for the latest in-window ERC-20 allowance, set by `approve` or `permit`, that the spender v holds from the owner u, valued in US dollars and weighted as AWP weights a transfer. The edge records a delegation of spending rights, not a history of payments. Approvals have been used to cluster the addresses of one entity (Victor, 2020), but no reputation method reviewed ranks on them (Section 2.21).
- Same-cohort evaluation with statistical inference: every score is computed on one set of contracts and checked against identical transfer, allowance and Sybil-stability proxies under the protocol used with AWP (T. Do et al., 2023), with 95% bootstrap intervals for the coefficients and paired intervals for the contrasts (Efron, 1979).
- Out-of-window tests with degree baselines: scores frozen at t_1 are tested against the new approvals and new transfer senders that arrive after t_1, beside the raw degrees that PageRank smooths, which separates in-sample agreement from prediction and the propagated score from

its local count (Claims C4 and C5). In contrasts computed after the labels were known, AWP beat its degree in both windows and EndorseRank did so only with AWP's activity restarts, a hypothesis for a registered test. A comparison registered with W1 adds an outcome built from neither edge type: on later liquidation avoidance the approval count showed no lead over the transfer count in the registered window, though it led in W0, and the like-for-like PageRank pair is not resolved, in small trader samples (Claim C6).

- Closed form of EndorseRank on a depth-one graph: with uniform restarts and uniformly redistributed dangling mass, a spender whose in-neighbors have no in-edges scores $c\,(1 + d\,\delta(v))$, where c is the common score of nodes without in-edges, d the damping factor and $\delta(v)$ the owner-normalized in-strength, which divides each owner's endorsement over all the spenders it approves. This holds to within 1% for every depth-one spender but for only a minority of the deeper ones, so multi-hop propagation is present (Section 5.3.2). Besides the time and token weights, that normalization and this propagation separate EndorseRank from the raw approval count, but over all spenders it did not rank later approvals above that count, and deeper spenders were not tested apart.
- Coupled operator and its two rules: C-PR blends a US-dollar allowance layer, with each allowance capped, and a US-dollar transfer layer node by node with a fixed weight λ, and S-PR is a seeded ablation. Rule A, fixed with λ in the analysis plan, asked C-PR in its primary part to do no worse than the allowance walk on new approval pairs and better than the transfer walk on new senders, and rule B, registered with W1, asked it to improve on the transfer walk; on the revised scores, in a re-analysis, both fail as in the registered analysis (Appendix 8.6), though the gain over the transfer walk on new approval pairs recurs in W1 (Claim C5). The ablation shows that the allowance signal survives when allowance edges carry the walk and is mostly lost when they only decide where it restarts.

1.10 Scope and limitations

The study covers Arbitrum One from 2023-10-01 to 2026-06-30, and W1 adds July to September 2026. Its main sample is the matched cohort of $n = 15{,}107$ contracts, and its holdout samples the contract spenders of each freeze ($n = 17{,}050$ in W0, $n = 18{,}152$ in W1). The scored wallets are contract accounts, because user wallets serve too many different purposes for their activity to be put in one order (Section 3.2); account type is read from the code at a pinned block, not at the time of each event. Only events of the five tokens with a cohort contract at one end enter the graphs, so each graph is the ego network of the cohort, and a multi-hop path is seen only where each of its edges touches a cohort contract (Section 5.3.2). Few GMX V2 contract accounts in the freeze-date graph closed at least three positions in these windows ($n = 88$ in W0, $n = 70$ in W1), so every trader result has wide intervals (Section 4.6.3). No claim is made that the results generalize beyond Arbitrum One or the months studied. Amounts are in US dollars, a choice made after the registered results were known (Section 3.5.1); EndorseRank and AWP pass them through a bounded function that gives the median amount half weight, while the layers of C-PR use the amounts themselves, with each allowance capped (Chapter 3). Adversarial manipulation of allowances or transfers is a real threat, because addresses are cheap to create (Douceur, 2002): Section 5.6 shows that a small farm of Sybil addresses can lift a contract's EndorseRank above most of the cohort cheaply, while reaching the top is expensive, and the Sybil-adjusted proxies describe some manipulation patterns without protecting the score against them.

1.11 Research roadmap

Chapter 2 reviews graph-based on-chain reputation, ERC-20 allowance semantics and the reputation tooling deployed in DeFi, and develops the theoretical framework. Chapter 3 defines the cohorts, the tokens, prices and the revision of the amounts, the graphs and scores, the construct checks, the temporal holdouts, the timing protocol, and the standards for reproducibility and ethics. Chapter 4 reports the benchmark (Claim C1), the same-window alignment of every score (C2),

rank divergence (C3), robustness checks, the temporal holdouts W0 and W1 (C4 and C5) and the comparison on liquidation avoidance (C6); Appendix 8.5 collects the direct comparisons between EndorseRank and AWP, and Appendix 8.6 reports the registered analysis. Chapter 5 interprets the findings by research question and objective, works out their implications, models the cost of Sybil attacks on allowance graphs, and weighs threats to validity and limitations. Chapter 6 answers RQ1–RQ5, states the contribution to knowledge and sets out future work.